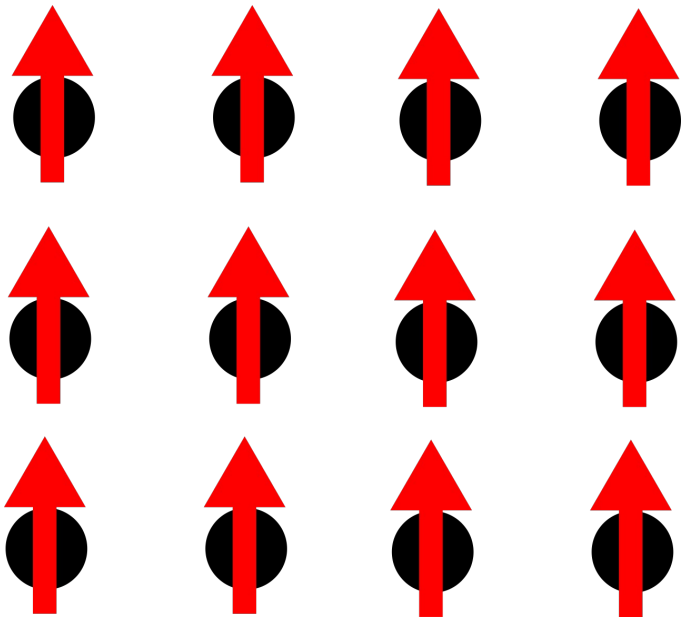
Quantum magnetism, spinons and Kondo physics

Today’s learning outcomes

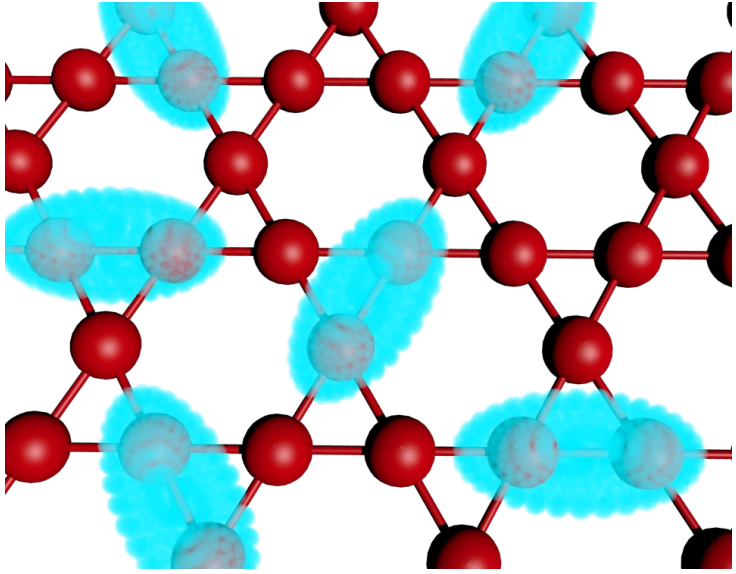
- Understand minimal models of quantum magnetism
- Identify the quantum excitations of quantum spin-liquids
- Identify the fundamental physics of Kondo lattice models

Classical and quantum magnetism

Ferromagnet & antiferromagnets

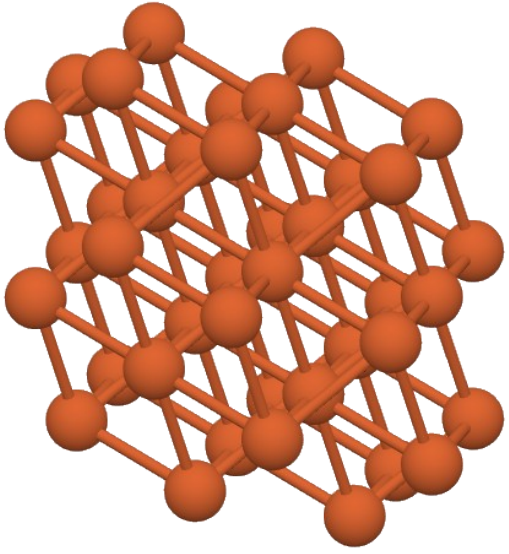


Quantum spin-liquids



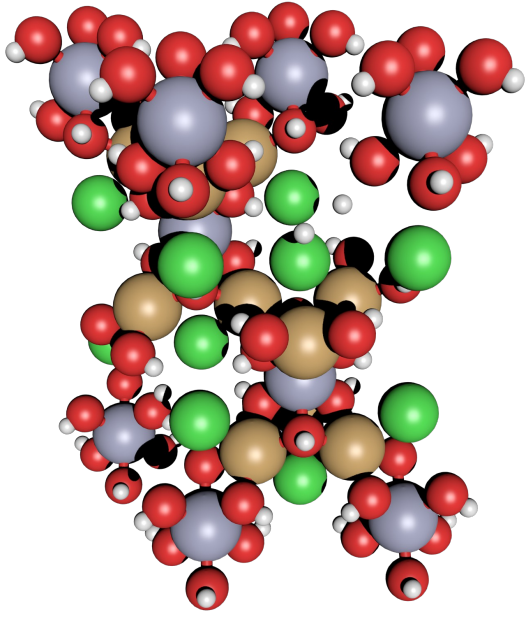
Classical and quantum magnetic materials

Ferromagnets & antiferromagnets



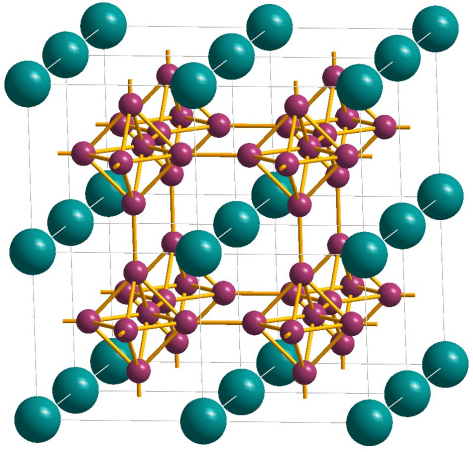
Iron

Quantum spin-liquids



Herbertsmithite

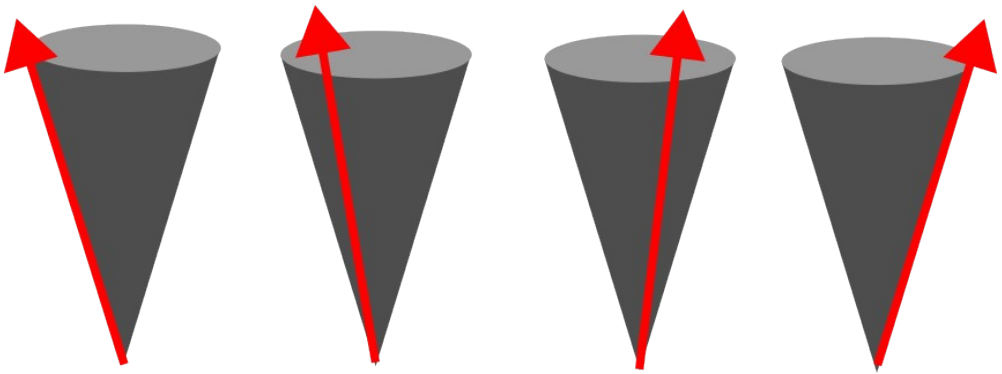
Heavy-fermion insulators



SmB_6

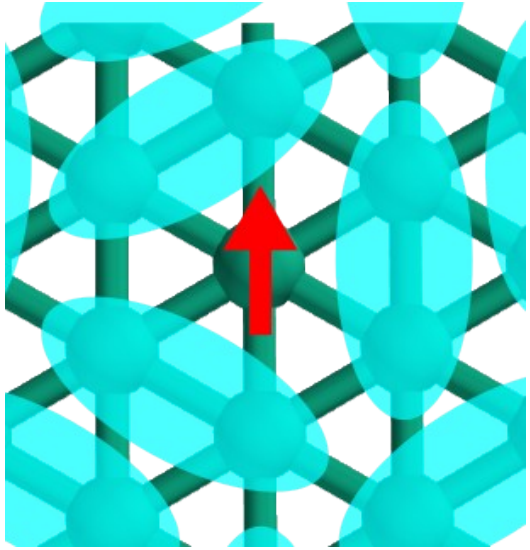
Two excitations in magnetic materials

Magnons



S=1
No charge

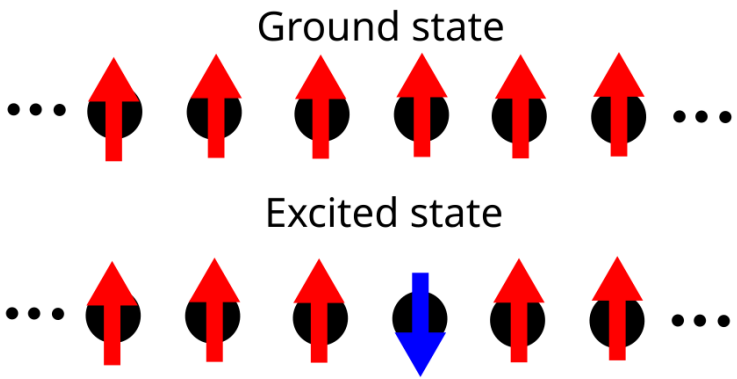
Spinons



S=1/2
No charge

Two excitations in magnetic materials

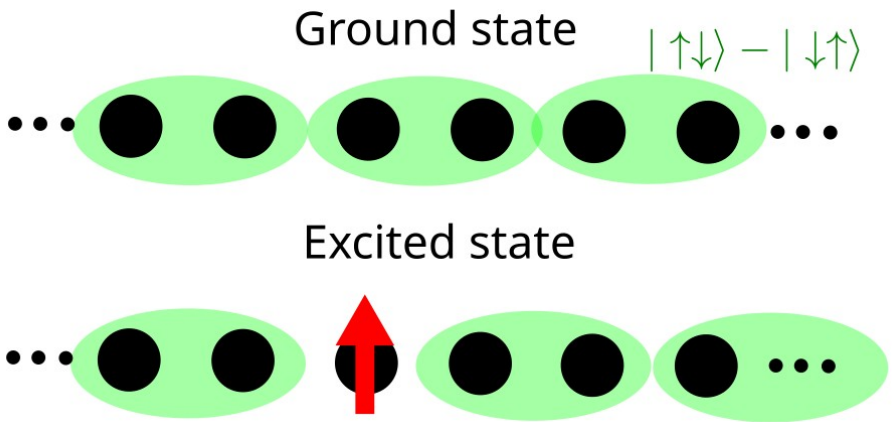
Excitations in a symmetry broken magnet



$$\Delta S = 1$$

Magnons

Excitations in a quantum spin liquid



$$\Delta S = 1/2$$

Spinons

Quantum magnetism and spinons

The Ising dimer

What is the ground state of this Hamiltonian

$$\mathcal{H} = S_0^z S_1^z$$

The Hamiltonian has two ground states (related by time-reversal symmetry)

$$|GS_1\rangle = |\uparrow\downarrow\rangle \qquad |GS_2\rangle = |\downarrow\uparrow\rangle$$

Each ground state breaks time-reversal symmetry

A symmetry broken antiferromagnet is a macroscopic version of this

The quantum Heisenberg dimer

What is the ground state of this quantum Hamiltonian?

$$\mathcal{H} = \vec{S}_0 \cdot \vec{S}_1$$

The ground state is unique, and does not break time-reversal

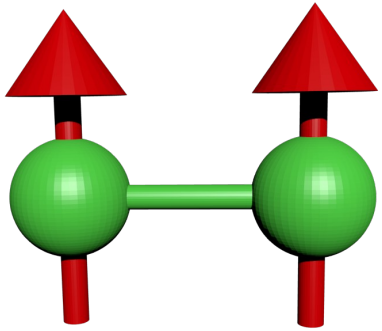
$$|GS\rangle = \frac{1}{\sqrt{2}}[|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle] \qquad \langle \vec{S}_i \rangle = 0$$

The state is maximally entangled

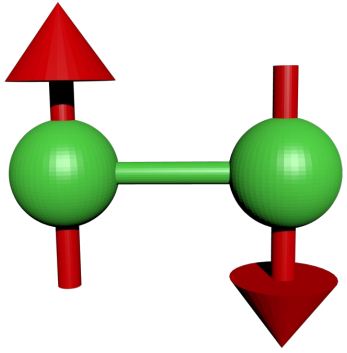
Can we have a macroscopic version of this ground state? $\langle \vec{S}_i \rangle = 0$

Towards quantum-spin liquids

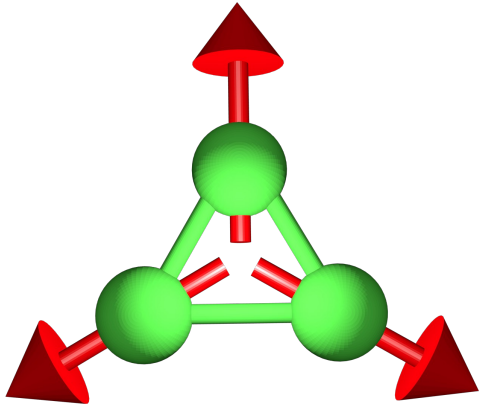
Ferromagnetism



Antiferromagnetism



Frustrated magnetism

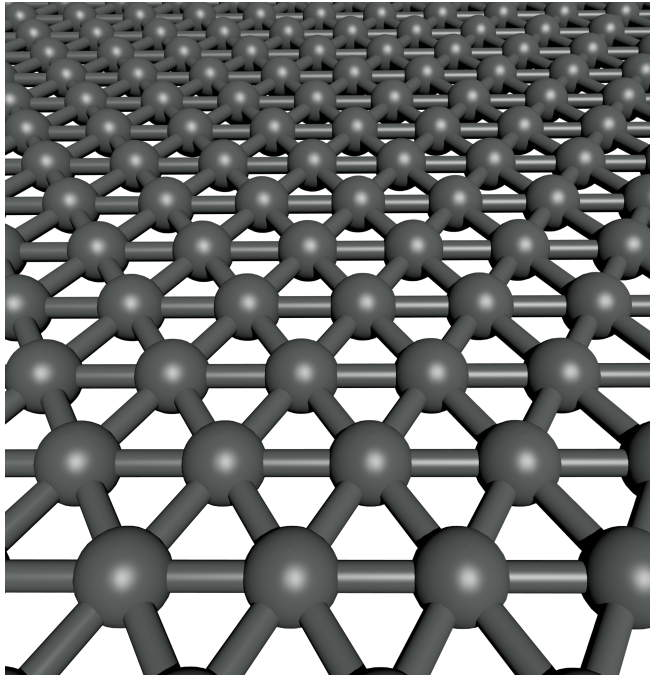


To get a quantum-spin liquid, we should look for frustrated magnetism

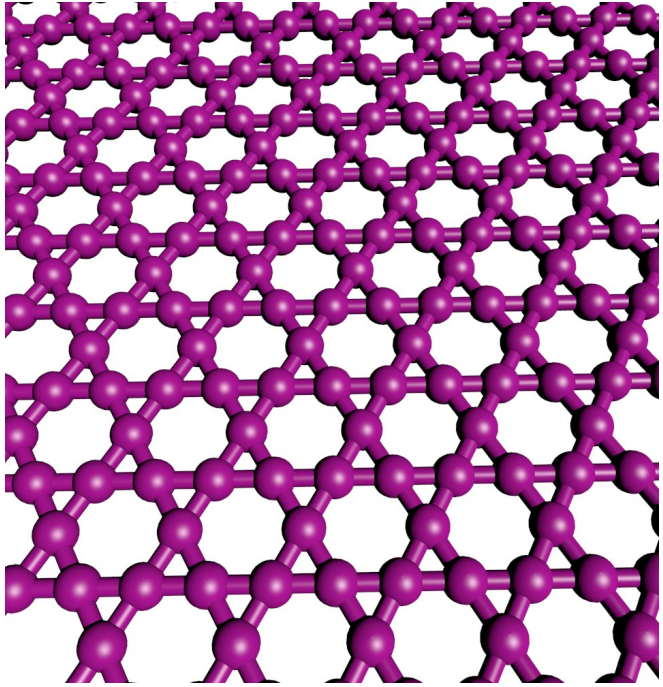
$$\langle \vec{S}_i \rangle = 0$$

Frustrated lattices

Triangular



Kagome



A minimal quantum magnet

Take a dimerized Heisenberg model of the form $H = \sum_n J \vec{S}_{2n} \cdot \vec{S}_{2n+1}$



The many-body wavefunction is just a product of singlet between dimers

$$|\Psi\rangle \sim \prod_{\otimes n} \left(|\uparrow_{2n}\downarrow_{2n+1}\rangle - |\downarrow_{2n}\uparrow_{2n+1}\rangle \right)$$

The ground state is time-reversal invariant $\langle \vec{S}_i \rangle = 0$

The ground state of a quantum spin liquid

Let us assume that a certain Hamiltonian realizes a QSL $\mathcal{H} = \sum_{ij} J_{ij} \vec{S}_i \cdot \vec{S}_j$

Typical ground state wavefunction is a combination of singlets

$$(i,j) = \frac{1}{\sqrt{2}}(|\uparrow_i\downarrow_j\rangle - |\downarrow_i\uparrow_j\rangle) \qquad |\Psi_{RVB}\rangle = \sum_{i_1 j_1 \cdots i_n j_n} a_{(i_1 j_1 \cdots i_n j_n)} \, |(i_1,j_1)\cdots(i_n,j_n)\rangle$$

A variational wavefunction can be written by projecting a BCS wavefunction

$$|\Psi_{RVB}\rangle = P_G \, |\Psi_{BCS}\rangle \,,$$
$$|\Psi_{BCS}\rangle = \prod_{\mathbf{k}} (u_{\mathbf{k}} + v_{\mathbf{k}} c^\dagger_{\mathbf{k}\uparrow} c^\dagger_{-\mathbf{k}\downarrow}) \, |0\rangle$$

Projector on singly occupied subspace P_G

Quasiparticles in a quantum spin-liquid

Let us assume that a certain Hamiltonian realizes a QSL $\mathcal{H} = \sum_{ij} J_{ij} \vec{S}_i \cdot \vec{S}_j$

Quantum spin liquids require $\langle \vec{S}_i \rangle = 0$

The approximation used for magnons breaks down

$$\langle S_i^z \rangle = S - \langle a_i^\dagger a_i \rangle$$

$$\langle a_i^\dagger a_i \rangle \ll S$$

We need a new approximation for the quantum excitations

The parton transformation

Transform spin operators to auxiliary fermions (Abrikosov fermions)

$$S_i^\alpha = \frac{1}{2}\sigma^\alpha_{s,s'} f^\dagger_{i,s} f_{i,s'}$$

The fermions f (spinons) have S=1/2 but no charge

This transformation artificially enlarges the Hilbert space, thus we have to put the constraint

$$\sum_s f^\dagger_{i,s} f_{i,s} = 1$$

This transformation allow to turn a spin Hamiltonian into a fermionic Hamiltonian

The spinon Hamiltonian

We can insert the auxiliary fermions $S_i^\alpha \sim \sigma_{s,s'}^\alpha f_{i,s}^\dagger f_{i,s'}$

And perform a mean-field in the auxiliary fermions (spinons)

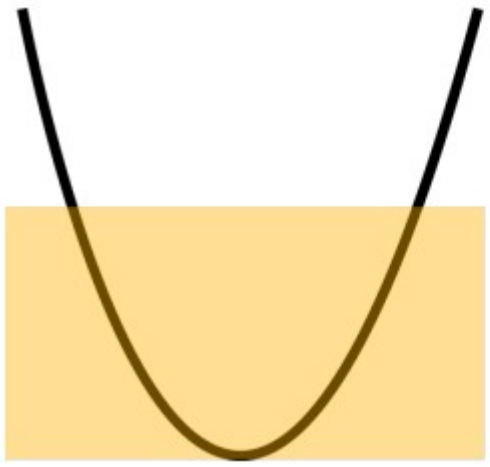
$$\mathcal{H} = \sum_{ij} J_{ij} \vec{S}_i \cdot \vec{S}_j \longrightarrow \mathcal{H} = \sum_{ij,s} \chi_{ij} f_{i,s}^\dagger f_{j,s}$$

Enforcing time-reversal symmetry $\langle \vec{S}_i \rangle = 0$

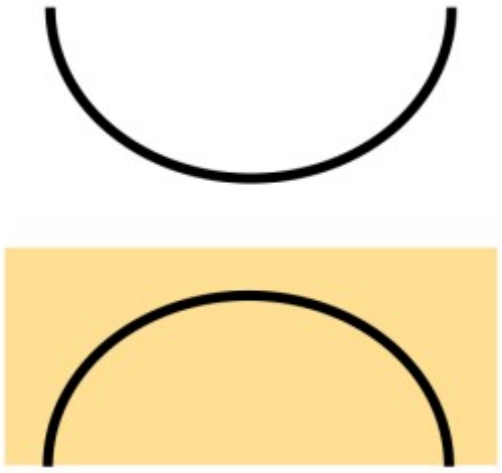
The exitations of the QSL are described by a single particle spinon Hamiltonian

Spinon dispersions

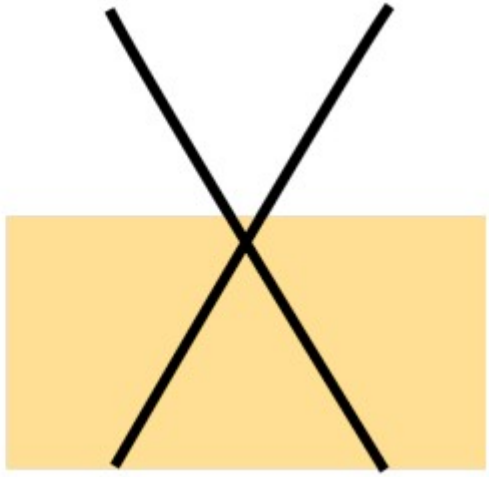
Gapless spinons



Gapped spinons



Dirac spinons



$$\mathcal{H} = \sum_{ij,s} \chi_{ij} f^{\dagger}_{i,s} f_{j,s}$$

Emergent gauge symmetry in quantum spin liquids

Gauge transformation in auxiliary fermions $f_{n,s} \to e^{i\theta_n} f_{n,s} \qquad f^\dagger_{n,s} \to e^{-i\theta_n} f^\dagger_{n,s}$

Hamiltonian and operators are invariant

$$\mathcal{H} = \sum_{ij} J_{ij} \vec{S}_i \cdot \vec{S}_j \to \mathcal{H}$$

$$S^\alpha_i = \frac{1}{2} \sigma^\alpha_{s,s'} f^\dagger_{i,s} f_{i,s'} \to S^\alpha_i$$

But mean-field auxiliary fermion Hamiltonian is not invariant

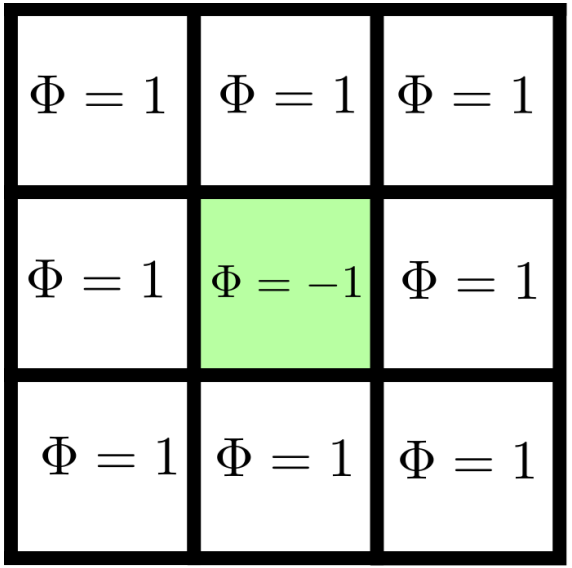
$$\mathcal{H} = \sum_{nm,s} \chi_{nm} f^\dagger_{n,s} f_{m,s} \to \sum_{nm,s} \chi_{nm} e^{i(\theta_n - \theta_m)} f^\dagger_{n,s} f_{m,s}$$

This leads to an emergent gauge theory for quantum spin liquids

Gauge excitations in quantum spin liquids

The spinon mean-field parameter is a dynamical degree that can have topological defects

$$\mathcal{H} = \sum_{ij,s} |\chi_{nm}| e^{i\theta_{nm}} f^{\dagger}_{n,s} f_{m,s}$$

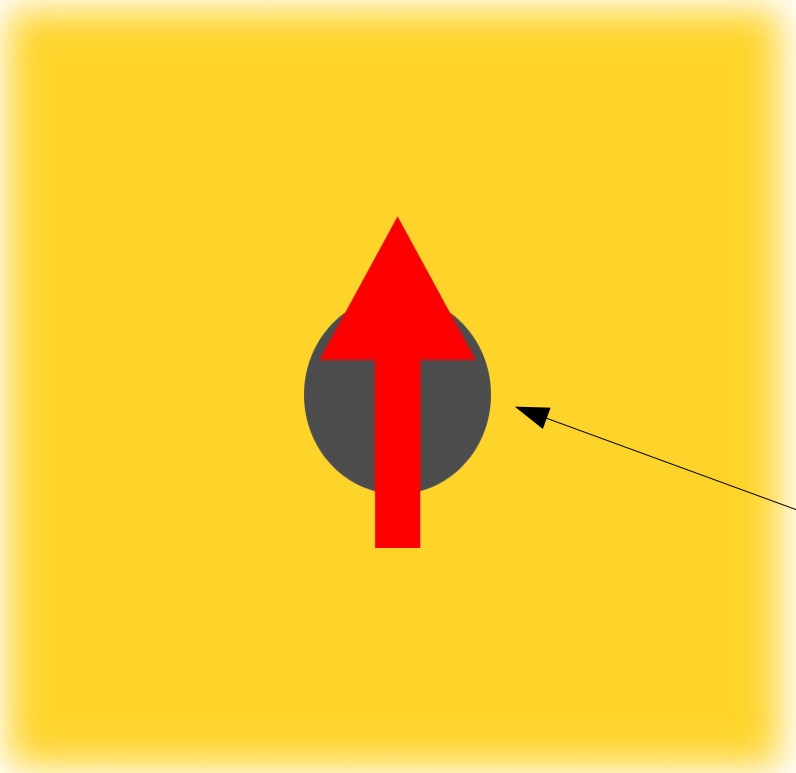


A topological defect on the gauge field realizes a gauge excitation (vison)

In general, gauge excitations are coupled to spinon excitations

The Kondo lattice model

The Kondo problem



Conduction electrons

$$H = -t \sum_{(i,j)\sigma} \left(c^{\dagger}_{i\sigma} c_{j\sigma} + \text{ H.c} \right)$$

Kondo coupling

$$H_K = \sum_{\alpha\beta} \left(c^{\dagger}_{0\beta} \vec{\sigma}_{\beta\alpha} c_{0\alpha} \right) \cdot \vec{S}$$

We now take a quantum spin S=1/2

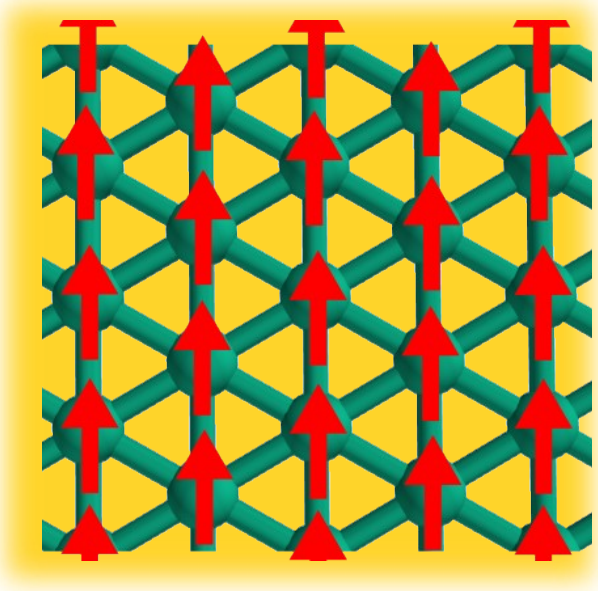
$$|GS\rangle \sim \frac{1}{\sqrt{2}} [|\Uparrow\downarrow\rangle - |\Downarrow\uparrow\rangle]$$

The Kondo lattice problem

The Kondo lattice problem

$$H = -t \sum_{(i,j)\sigma} \left(c^{\dagger}_{i\sigma} c_{j\sigma} + \text{ H.c } \right) + J \sum_{j,\alpha\beta} \left(c^{\dagger}_{j\beta} \vec{\sigma}_{\beta\alpha} c_{j\alpha} \right) \cdot \vec{S}_j$$

Conduction electrons



Kondo coupling

Kondo sites

Solving the Kondo lattice problem

$$H = -t \sum_{(i,j)\sigma} \left(c^{\dagger}_{i\sigma} c_{j\sigma} + \text{ H.c } \right) + J \sum_{j,\alpha\beta} \left(c^{\dagger}_{j\beta} \vec{\sigma}_{\beta\alpha} c_{j\alpha} \right) \cdot \vec{S}_j$$

Replace the spin sites by auxiliary fermions

$$S_{\alpha\beta}(j) = f^{\dagger}_{j\alpha} f_{j\beta} - \frac{n_f(j)}{N} \delta_{\alpha\beta}$$

This makes the effective Hamiltonian an “interacting” fermionic Hamiltonian

$$H \sim \sum_{\mathbf{k}\alpha} \epsilon_{\mathbf{k}} c^{\dagger}_{\mathbf{k}\alpha} c_{\mathbf{k}\alpha} - J \sum_{j,\alpha\beta} \left(c^{\dagger}_{j\beta} f_{j\beta} \right) \left(f^{\dagger}_{j\alpha} c_{j\alpha} \right)$$

Solving the Kondo lattice problem

Now we decouple the fermions with a mean-field approximation

$$H \sim \sum_{\mathbf{k}\alpha} \epsilon_{\mathbf{k}} c^{\dagger}_{\mathbf{k}\alpha} c_{\mathbf{k}\alpha} - J \sum_{j,\alpha\beta} \left(c^{\dagger}_{j\beta} f_{j\beta} \right) \left(f^{\dagger}_{j\alpha} c_{j\alpha} \right)$$

Obtaining a quadratic Hamiltonian

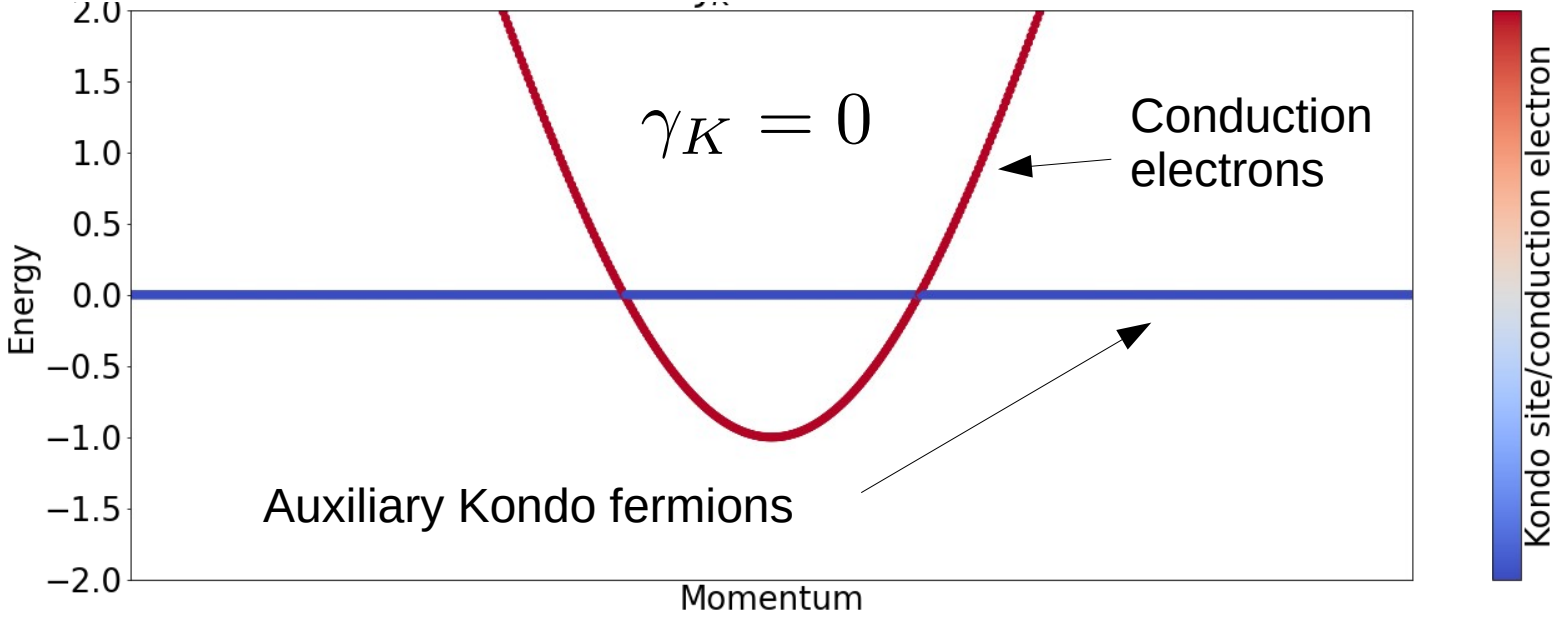
$$H \sim \sum_{\mathbf{k}\alpha} \epsilon_{\mathbf{k}} c^{\dagger}_{\mathbf{k}\alpha} c_{\mathbf{k}\alpha} - \gamma_K \sum_{\mathbf{k},\alpha\beta} f^{\dagger}_{\mathbf{k}\alpha} c_{\mathbf{k}\alpha} + h.c.$$

Conduction band dispersion

Kondo hybridization

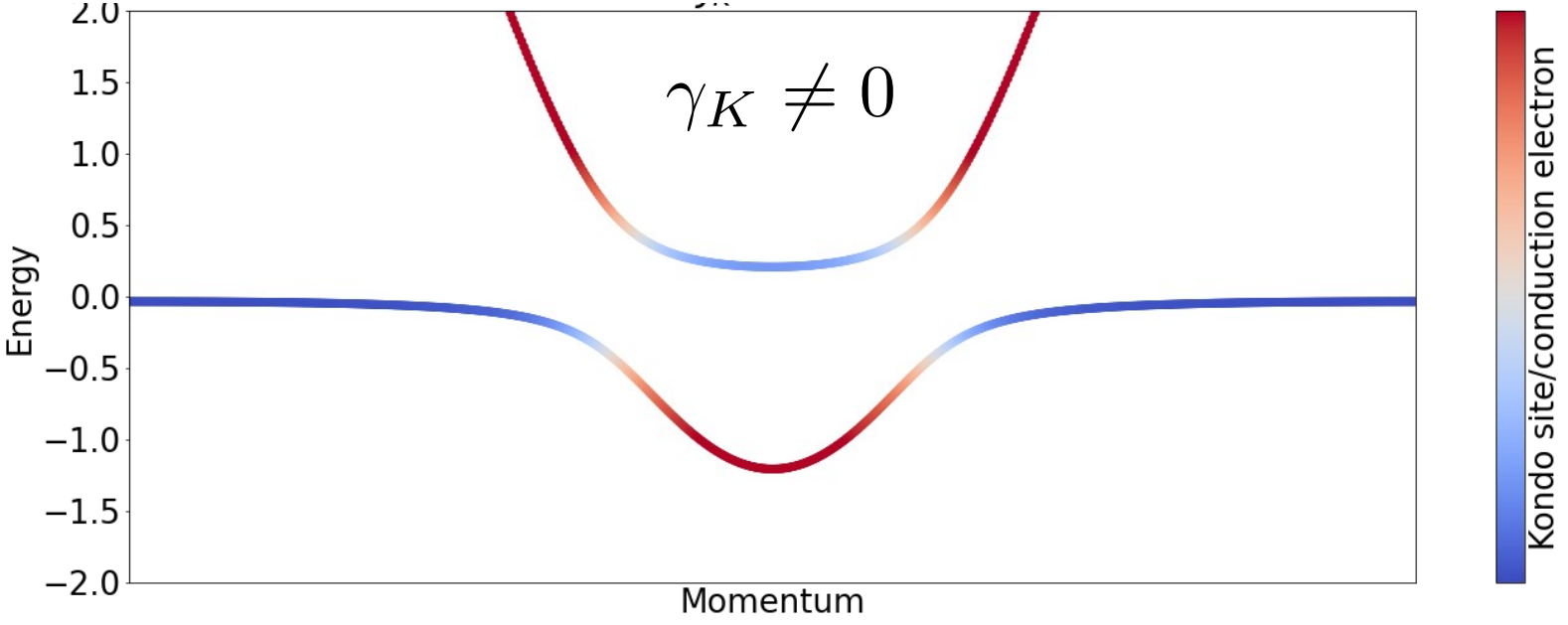
Electronic structure of the Kondo lattice problem

$$H \sim \sum_{\mathbf{k}\alpha} \epsilon_{\mathbf{k}} c^{\dagger}_{\mathbf{k}\alpha} c_{\mathbf{k}\alpha} - \gamma_K \sum_{\mathbf{k},\alpha\beta} f^{\dagger}_{\mathbf{k}\alpha} c_{\mathbf{k}\alpha} + h.c.$$



Electronic structure of the Kondo lattice problem

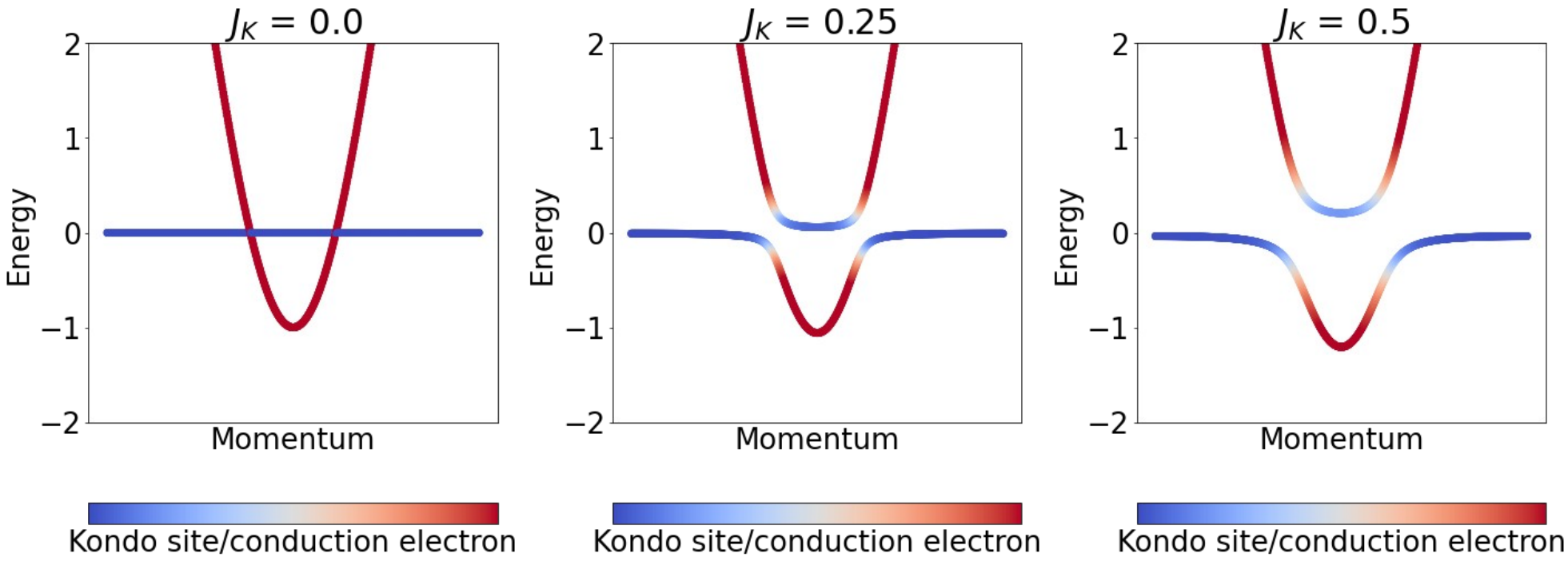
$$H \sim \sum_{\mathbf{k}\alpha} \epsilon_{\mathbf{k}} c^{\dagger}_{\mathbf{k}\alpha} c_{\mathbf{k}\alpha} - \gamma_K \sum_{\mathbf{k},\alpha\beta} f^{\dagger}_{\mathbf{k}\alpha} c_{\mathbf{k}\alpha} + h.c.$$



The Kondo coupling opens up a gap in the electronic structure

Dependence on the Kondo coupling

The heavy-fermion gap becomes bigger as the Kondo coupling increases



Take home

- Frustrated magnetic models can display quantum spin-liquid behavior
- The fundamental excitations of QSL have S=1/2
- Quantum entanglement can appear between spins and electrons, giving rise to Kondo states